Lecture 29 (B28): Deep Uncertainty Quantification and Policy

Constrained Optimal Carbon Tax in a Stochastic OLG-IAM

Simon Scheidegger

HEC, University of Lausanne

Based on: Kübler, Scheidegger, Surbek (2025, arXiv:2507.01704)

Reading: `JPE_Macro.pdf`; public draft: arXiv:2507.01704

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Outline of This Lecture (∼75 min)

Part I: Motivation (∼10 min)

- Why OLG for climate policy?
- Winners and losers from carbon taxation
- Our contribution and the 3-step method

Part II: The OLG-IAM Model (∼15 min)

- Technology, households, climate
- Carbon tax and transfers
- The planner's problem

Part III: Full Derivation (∼20 min)

- Firm FOCs, household Euler equations
- Market clearing and equilibrium system
- Nonstationarity and time as state variable

Part IV: 3-Step Pipeline (∼15 min)

Part V: Results (∼13 min)

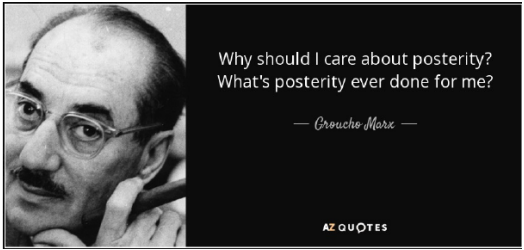
Bridge from Lecture 08

From RA → heterogeneous agents (OLG) and UQ → optimal policy search.

Part I

Optimal Policy with Heterogeneous Agents

The Challenge: Climate Risk and Heterogeneity



- Current research in climate change macro often resorts to simplifying assumptions (e.g., representative agent) or searches for Pareto improvements by guessing.
- We need tools for optimal policy within globally solved models that take into account heterogeneity and stochasticity.

Why Overlapping Generations (OLG)?

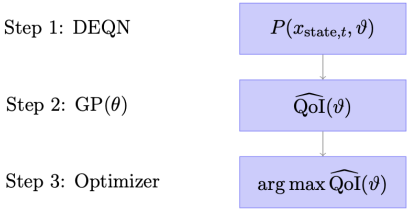
- **Intergenerational equity**: Climate policy creates winners and losers across generations.
- **Political feasibility**: Current cohorts bear costs; future cohorts reap benefits.
- **No representative agent**: With OLG, there is no single welfare function — must aggregate across cohorts.
- **Pareto constraint**: A policy that improves aggregate welfare but hurts current generations may be politically infeasible.

Key question

Can we find carbon tax rules that are Pareto-improving — i.e., no generation is worse off?

Our Contribution: A Scalable Policy Framework

The Method



A 3-step ML pipeline to alleviate the curse of dimensionality.

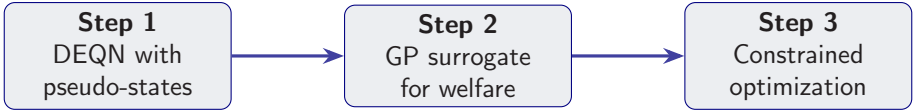
The Application

We compute the first globally optimal, Pareto-improving carbon tax in a stochastic OLG model with tipping points.

A Generic Blueprint

A universal framework for **almost any** stochastic heterogeneous-agent model: solve global equilibria and derive constrained-optimal policies.

The 3-Step Method Overview



- **Step 1**: Solve the OLG equilibrium globally for all tax rules at once (pseudo-states).
- **Step 2**: Build a GP surrogate mapping tax parameters $\rightarrow$ welfare.
- **Step 3**: Find optimal tax-transfer rule subject to Pareto constraints.

Part II

The Stochastic OLG-IAM Model

Model Overview

- **12 overlapping generations** of selfish agents.
- **5-year periods**: agents live from age 20 to age 80.
- Use a **simple climate module** (Dietz and Venmans, 2019): cumulative emissions $\rightarrow$ temperature.
- Use a **damage function with stochastic tipping** (Kotlikoff et al., 2021).
- **Exogenously declining carbon intensity** as in DICE-2016 (Nordhaus, 2017).
- Carbon tax revenues recycled through **parameterized transfer rules**.

Technology / Firms

Firms produce final output Y_t and emissions e_t with capital K_t and labor L_t, choosing abatement μ_t:

$$(Y_t,\ e_t) = \Big(\Omega_t(T_t)\Phi(\mu_t)K_t^{\alpha}L_t^{1-\alpha} + (1-\delta)K_t,\ \ \kappa_t(1-\mu_t)K_t^{\alpha}L_t^{1-\alpha}\Big)$$

- $\Omega_t(T_t)$: retained-output factor after climate damages
- $\Phi(\mu_t) = 1 - \theta_1\mu_t^{\theta_2}$: net-of-abatement-cost factor
- κ_t: carbon intensity (declines stochastically over time)

Profit maximization with carbon tax τ_t:

$$\max_{\mu_t, K_t, L_t}\ \ \Omega_t(T_t)(1-\theta_1\mu_t^{\theta_2})K_t^{\alpha}L_t^{1-\alpha} + (1-\delta)K_t - w_tL_t - (1+r_t)K_t - \tau_t(1-\mu_t)\kappa_tK_t^{\alpha}L_t^{1-\alpha}$$

subject to $0 \le \mu_t \le 1$.

Households

Agents enter the labor force at age 20 with 0 assets and die at age 80 (12 periods).

Lifetime utility:

$$\mathbb{E}_t \sum_{j=1}^{12} \beta^{j-1} \frac{C_{t+j-1,j}^{1-\sigma_u}}{1-\sigma_u}$$

Budget constraint:

$$C_{t,j} + a_{t+1,j+1} = (1+r_t)a_{t,j} + w_t l_j + \mathbb{T}_{t,j}$$

- $C_{t,j}$, $a_{t,j}$, l_j, $\mathbb{T}_{t,j}$: consumption, assets, labor endowment, transfer of generation born at t, age j
- Agents work for 8 periods, then retire (ages 9–12: 40% labor as pension proxy)
- β: discount factor, σ_u: coefficient of relative risk aversion

Welfare criterion: $U_t = \mathbb{E}_0 \sum_{j=1}^{12} \beta^{j-1} \frac{C_{t+j-\mathbf{1},j}^{\mathbf{1}-\sigma_u}}{1-\sigma_u}$; use $\tilde{U}_t$ under taxation.

Carbon Tax and Transfer Schemes

- Tax τ_t on emissions; revenue redistributed to currently alive agents each period.
- Transfer schemes may create winners and losers (Kotlikoff et al., 2021).

Two transfer policies:

1. "10% declining scheme": youngest generation gets the highest share, each older generation gets 10% less:

$$\mathbb{T}_{t,j} = 0.9^{(j-1)} \cdot 0.1394 \cdot \tau_t e_t, \quad j \in \{2, \ldots, 12\}$$

2. "Optimal transfers": transfer shares to each generation are optimized (fixed at $t = 0$).

Normalization: shares sum to 1.

Climate Module: Cumulative Emissions → Temperature

Climate science shows a near-linear link between cumulative CO_2 emissions and temperature:

$$T_t^{\mathrm{AT}} \approx \sigma_{\mathrm{CCR}} \sum_{s=0}^{t} E_s$$

where σ_{CCR} is the Carbon-Climate Response (TCRE), ranging from 1.0°C to 2.3°C per 1000 GtC.

Damage function with stochastic tipping:

$$\Omega_t(T_t) = \frac{1}{1 + \left(\frac{1}{\psi_1} T_t\right)^2 + \left(\frac{1}{2 \cdot TP_t} T_t\right)^{6.754}}$$

TP_t follows a truncated random walk in $[2.5, 3.5]$.

Stochastic shocks:

- κ_t: AR(1) with time-varying persistence

$$\kappa_t = \rho_t \kappa_{t-1} + \epsilon_{\kappa,t}$$

$$\rho_t = \rho_0 + (\rho_\infty - \rho_0)(1 - e^{-\delta^\rho 5t})$$

- TP_t: random walk

$$TP_t = TP_{t-1} + \epsilon_{TP,t}$$

- Both shocks: 3-point discretization

PAUSE — 5 min Finger Exercise: Interpreting the Social Cost of Carbon

Exercise (5 min)

Suppose the damage-loss function is $D(T) = 1 - \frac{1}{1+\pi_1 T^2}$ with $\pi_1 = 0.00236$ and current warming $T = 1.1^\circ\mathrm{C}$.

1. Compute the current output loss $D(1.1)$ as a percentage of GDP.
2. If global GDP is \$100 trillion, what is the annual damage in dollars?
3. The SCC is defined as the present-value damage from one additional ton of CO_2. If the SCC is \$50/tCO_2: should a firm that can abate at \$30/tCO_2 do so? What about at \$80/tCO_2?

Solution: Interpreting the Social Cost of Carbon

Answer

1. $D(1.1) = 1 - \frac{1}{1+0.00236\times 1.21} = 1 - \frac{1}{1.002856} \approx 0.00285 = 0.285\%$ of GDP.
2. 0.00285×100 trillion $\approx$ \$285 billion per year.
3. Abate at \$30/tCO_2: yes—the cost (\$30) is below the social benefit (\$50). At \$80/tCO_2: no—the cost exceeds the benefit. The SCC is the optimal carbon tax: abate all sources cheaper than SCC, leave the rest.

The Planner's Problem

Objective: Choose policy parameters ϑ (taxes & transfers) to maximize social welfare:

$$\max_{\vartheta} \ \mathcal{W}(\vartheta) = \sum_{t=-10}^{30} \gamma_t \tilde{U}_t(\vartheta) \quad \text{s.t.} \quad \tilde{U}_t(\vartheta) \geq U_t \ \ \forall t \ \ \text{(if Pareto constrained)}$$

Welfare metric: Consumption equivalence factor

$$\mathrm{CE}_t = \left(\frac{\tilde{U}_t}{U_t} \right)^{\frac{1}{1-\sigma_u}} - 1 \qquad \text{(where } U_t \text{ is BAU utility)}$$

Scope:

- Horizon: 40 generations (11 current, 29 future).
- Complexity: high-dimensional ϑ in a stochastic setting.
- Pareto constraint: every cohort weakly better off than under BAU.
- Generation weights $\gamma_t = 1$ (utilitarian; Negishi or discounted weights also admissible).

Part III

Full Derivation — Equilibrium Conditions

Firm's Optimization Problem

Firms maximize profits given a carbon tax τ_t:

$$\max_{\mu_t, K_t, L_t} \quad \Omega_t(T_t)(1 - \theta_1 \mu_t^{\theta_2}) K_t^{\alpha} L_t^{1-\alpha} + (1-\delta)K_t - w_t L_t - (1+r_t)K_t - \tau_t(1-\mu_t)\kappa_t K_t^{\alpha} L_t^{1-\alpha}$$

subject to $0 \leq \mu_t \leq 1$.

Interpretation:

- Output net of damages and abatement costs
- Minus factor payments (wages, rental rate)
- Minus carbon tax on unabated emissions

Derivation: Firm First-Order Conditions

w.r.t. μ_t (marginal abatement cost = tax):

$$\tau_t \kappa_t = \Omega_t(T_{\mathrm{AT},t})\, \theta_1 \theta_2\, \mu_t^{\theta_2 - 1}$$

w.r.t. K_t (rental rate of capital):

$$r_t = \left[\Omega_t(T_{\mathrm{AT},t})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t (1 - \mu_t)\right] \alpha K_t^{\alpha-1} L_t^{1-\alpha} - \delta$$

w.r.t. L_t (wage rate):

$$w_t = \left[\Omega_t(T_{\mathrm{AT},t})(1 - \theta_1 \mu_t^{\theta_2}) - \tau_t \kappa_t (1 - \mu_t)\right] (1 - \alpha) K_t^{\alpha} L_t^{-\alpha}$$

Intuition

- Abatement FOC: firm abates up to the point where marginal cost equals the tax
- Factor prices: net of both damage and tax wedge on emissions

Household's Optimization Problem

Household born at time t solves:

$$\max_{\{C_{t+j-1,j},\, a_{t+j,j+1}\}_{j=1}^{12}} \quad \mathbb{E}_t \sum_{j=1}^{12} \beta^{j-1} \frac{C_{t+j-1,j}^{1-\sigma_u}}{1-\sigma_u}$$

Subject to:

$$C_{t,j} + a_{t+1,j+1} = (1+r_t)a_{t,j} + w_t l_j + \mathbb{T}_{t,j} \qquad \forall j = 1, \ldots, 12$$

Boundary conditions:

$$a_{t,1} = 0 \qquad \text{(born with no assets)}, \qquad a_{t+1,12} = 0 \qquad \text{(no bequest)}$$

Write the Lagrangian with multipliers $\lambda_{t,j}$ for each period's budget constraint.

Derivation: Household Euler Equation

FOC w.r.t. $C_{t,j}$:

$$\beta^{j-1} C_{t,j}^{-\sigma_u} = \lambda_{t,j}$$

FOC w.r.t. $a_{t+1,j+1}$:

$$\lambda_{t,j} = \mathbb{E}_t \left[(1 + r_{t+1}) \lambda_{t+1,j+1} \right]$$

Combining — the Euler equation:

$$\boxed{C_{t,j}^{-\sigma_u} = \beta \, \mathbb{E}_t \left[(1 + r_{t+1}) \, C_{t+1,j+1}^{-\sigma_u} \right] \qquad \forall j = 1, \ldots, 11}$$

Plus budget constraint:

$$C_{t,j} + a_{t+1,j+1} = (1 + r_t) a_{t,j} + w_t l_j + \mathbb{T}_{t,j}$$

Note

11 Euler equations + 12 budget constraints + 2 boundary conditions per cohort.

Market Clearing Conditions

Capital market:

$$\sum_{j=1}^{12} a_{t,j} = K_t \qquad \forall t$$

Labor market:

$$\sum_{j=1}^{12} l_j = L_t \qquad \forall t$$

Government budget (tax revenue = transfers):

$$\sum_{j=1}^{12} \mathbb{T}_{t,j} = \tau_t \, e_t \qquad \forall t$$

Complete system

Firm FOCs + household Euler equations + market clearing + climate laws of motion = full equilibrium.

Nonstationarity and Time as State Variable (Reprise)

Same lesson as Lecture 08 — now in OLG

The OLG-IAM is nonstationary: carbon intensity κ_t, abatement cost $\theta_{1,t}$, population, and TFP are all time-indexed exogenous sequences.

Specifics:

- κ_t: AR(1) with time-dependent coefficient $\rho_t = \rho_0 + (\rho_\infty - \rho_0)(1 - \exp(-\delta^\rho \cdot 5t))$
- Abatement cost $\theta_{1,t}$ declines over time
- Population and TFP are exogenous sequences

State vector:

$$\boldsymbol{x} = \Big[\underbrace{t,\ TP_t,\ TP_{\text{reached}},\ \kappa_t,\ E_t}_{\text{5 non-asset states}},\ \underbrace{a_{t,j}}_{\text{12 assets}},\ \vartheta\Big] \in \mathbb{R}^{12+5+d_\vartheta}$$

Time enters the neural net as an input — the policy function is $\mathcal{N}_{\boldsymbol{\rho}}(t, \text{states}, \text{params})$.

Complete Equilibrium System Summary

Firm FOCs:

$$0 = \tau_t \kappa_t - \Omega_t \theta_{\mathbf{1}} \theta_{\mathbf{2}} \mu_t^{\theta_{\mathbf{2}} - \mathbf{1}} \qquad (\mu_t)$$

$$r_t = [\Omega_t (1 - \theta_{\mathbf{1}} \mu_t^{\theta_{\mathbf{2}}}) - \tau_t \kappa_t (1 - \mu_t)] \alpha K_t^{\alpha - \mathbf{1}} L_t^{\mathbf{1} - \alpha} - \delta \qquad (K_t)$$

$$w_t = [\Omega_t (1 - \theta_{\mathbf{1}} \mu_t^{\theta_{\mathbf{2}}}) - \tau_t \kappa_t (1 - \mu_t)](1 - \alpha) K_t^{\alpha} L_t^{-\alpha} \qquad (L_t)$$

Household Euler equations:
For all $j = 1, \ldots, 11$:

$$C_{t,j}^{-\sigma_u} = \beta \, \mathbb{E}_t[(1 + r_{t+\mathbf{1}}) C_{t+\mathbf{1},j+\mathbf{1}}^{-\sigma_u}] \qquad \text{(EE)}$$

$$C_{t,j} + a_{t+\mathbf{1},j+\mathbf{1}} = (1 + r_t) a_{t,j} + w_t l_j + \mathbb{T}_{t,j} \qquad \text{(BC)}$$

Market clearing:

$$K_t = \sum_{j=\mathbf{1}}^{\mathbf{12}} a_{t,j} \qquad (K)$$

$$L_t = \sum_{j=\mathbf{1}}^{\mathbf{12}} l_j \qquad (L)$$

$$\tau_t e_t = \sum_{j=\mathbf{1}}^{\mathbf{12}} \mathbb{T}_{t,j} \qquad (\mathbb{T})$$

Climate:

$$T_t^{\mathrm{AT}} = \sigma_{\mathrm{CCR}} \sum_{s=\mathbf{0}}^{t} E_s$$

$$E_t = E_{t-\mathbf{1}} + \kappa_t (1 - \mu_t) K_t^{\alpha} L_t^{\mathbf{1} - \alpha}$$

Stochastic dynamics:

$$\kappa_t = \rho_t \kappa_{t-\mathbf{1}} + \epsilon_{\kappa,t}$$

$$TP_t = TP_{t-\mathbf{1}} + \epsilon_{TP,t}$$

From Equilibrium Conditions $\rightarrow$ DEQN Residuals

Each equation becomes a residual G_i; loss $= \sum_i G_i^2$.

Network inputs (state vector):

$$\boldsymbol{x} = [t,\ TP_t,\ TP_{\text{reached}},\ \kappa_t,\ E_t,\ a_{t,j},\ \vartheta] \in \mathbb{R}^{12+5+d_\vartheta}$$

Network outputs (policy vector):

$$y = [a_{t+1,j+1},\ v_{t,j}] \in \mathbb{R}^{2(A-1)} = \mathbb{R}^{22}$$

Why include ϑ?

- Tax rule parameters $\tau(\vartheta)$
- Transfer shares as pseudo-states
- DEQN learns policies for all tax rules and transfers at once

Diagnostics to report:

- Runtime and hardware configuration
- Residual MSE and Euler-error distribution
- Robustness to seeds and stress states

Part IV

The 3-Step Solution Pipeline

Step 1: DEQN with Pseudo-States

- Include tax parameters $\vartheta = (\vartheta_0, \vartheta_E, \vartheta_\kappa, \vartheta_{TP}) + 11$ transfer shares as pseudo-states.
- DEQN learns the equilibrium for all tax rules at once.

Hyperparameter	**Value**
Hidden layers	2
Units per layer	256
Activation	GeLU
Optimizer (LR)	Adam (10^{-5})
Batch size	64
Inputs	$12 + 5 + d_\vartheta$
Outputs	$2 \times (A - 1) = 22$
Sim. episode length	70 steps ($\times$ 5y = 350y)

Key advantage over brute-force

Pseudo-states replace many separate tax-rule solves with one amortized equilibrium solve.

Step 2: GP Surrogate for Welfare

Map $\vartheta \to \mathbb{E}[\mathcal{W}(\vartheta)]$ via Gaussian Process:

- 450 LHS points + 50 BAL points = 500 total design points
- Leave-one-out error $\sim 10^{-4}$
- Two targets:
  1. Aggregate welfare scalar $\mathcal{W}(\vartheta)$
  2. Vector of 40 cohort utilities $\tilde{U}_t(\vartheta)$ (for Pareto constraints)

Why a surrogate?

- Each DEQN simulation is fast ($\sim$ seconds), but evaluating thousands of ϑ for optimization is still expensive.
- GP provides predictive uncertainty $\to$ guides BAL acquisition.
- 21st-century lookup table: Welfare$(\vartheta_1, \ldots, \vartheta_N)$.

Step 3: Constrained Optimization

Find optimal tax-transfer rule:

$$\vartheta^* = \arg\max_{\vartheta} \mathcal{W}(\vartheta) \qquad \text{s.t.} \quad \tilde{U}_t(\vartheta) \geq U_t \;\; \forall t$$

- Linear constraints on tax function staying in feasible range
- Pareto constraints ensure no generation is worse off
- Optimization runs on the GP surrogate (milliseconds per evaluation)

Computational savings

Full constrained policy search in minutes, not days/weeks.

Three Policy Scenarios

1. **Scenario 1: Unconstrained welfare max with exogenous transfers**
   - Tax rule: $\tau_t(E_t) = \vartheta_0 + \vartheta_E \cdot E_t$
   - 10% declining transfer scheme
2. **Scenario 2: Pareto-improving with optimal transfers**
   - Same tax rule + endogenous $\mathbb{T}_{t,j}$
   - Pareto constraint: $\tilde{U}_t \geq U_t$ for all cohorts
3. **Scenario 3: Richer tax rule**
   - $\tau_t = \vartheta_0 + \vartheta_E \cdot E_t + \vartheta_\kappa \cdot \kappa_t + \vartheta_{TP} \cdot (1 - D_{TP})$
   - Incorporates carbon intensity and tipping-proximity signals

Why linear rules? Interpretable, implementable, and transparent for policy communication.

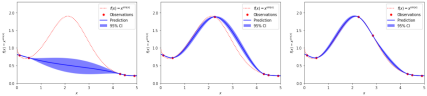
Bayesian Active Learning for Welfare Surrogate

Score function:

- Combine GP predictive variance with proximity to optimum
- Explore-exploit tradeoff in policy space
- Add points where uncertainty is highest or welfare looks promising

Convergence:

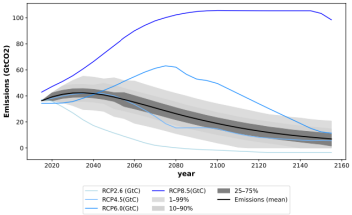
- LOO error drops rapidly with BAL points
- 500 points sufficient for stable policy conclusions



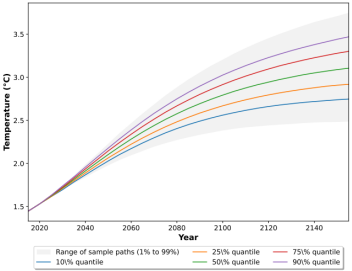
Part V

Results

Business-As-Usual (BAU) Scenario



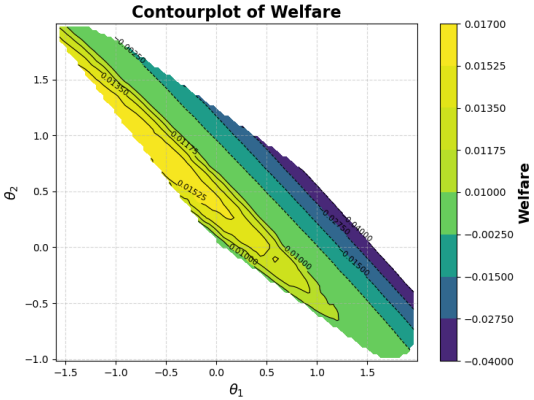
(a) Emissions



(b) Temperature

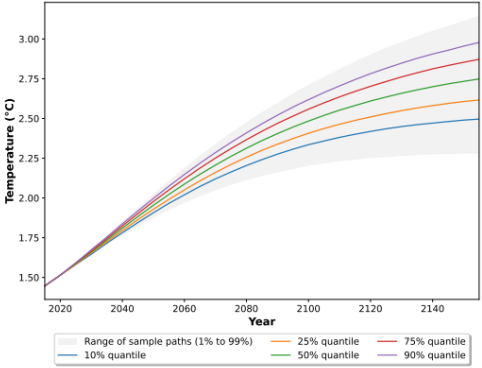
Emissions and temperature paths under no climate policy, compared with RCP scenarios.

Welfare Surrogate Surface (Scenario 1)



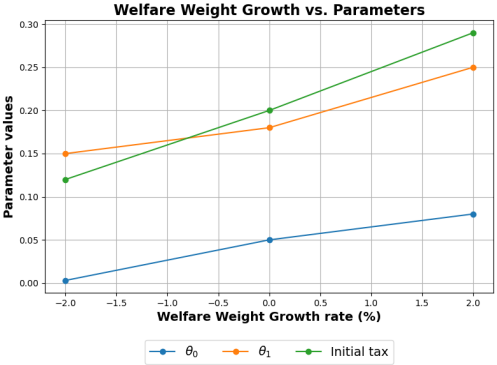
- Objective surface in $(\vartheta_0, \vartheta_E)$ from GP surrogate.
- Smooth structure enables efficient constrained optimization.
- The unconstrained optimum is easy to identify but is not Pareto-feasible.

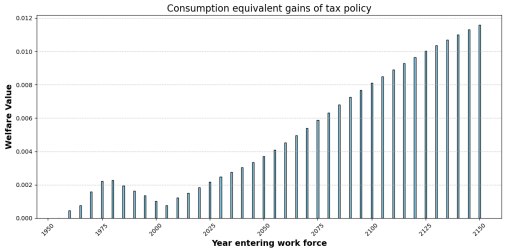
Scenario 1: Unconstrained Optimal Tax



- Strong emission reduction.
- Significant warming reduction.
- But: creates losers among current cohorts.
- Welfare losses up to ∼5% for transition generations.
- Politically fragile without compensation.

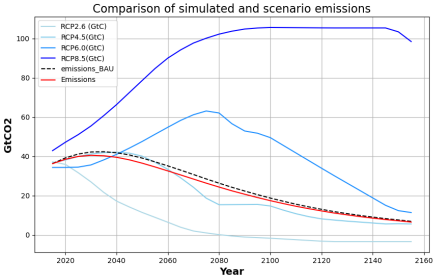
Scenario 2: Pareto-Improving Policy





Left: Pareto frontier in $(\vartheta_0, \vartheta_E)$-space. Right: welfare gains by cohort — all non-negative under Pareto-improving policy.

Pareto Policy as Emissions-Tail Insurance

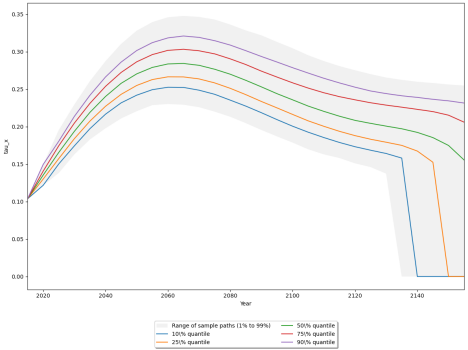


- Mean emissions move moderately.
- Upper-tail emissions are compressed materially.
- Main mechanism: disaster-risk reduction (tail insurance), not maximal average abatement.

Insight

Transfers preserve non-negative welfare effects for transition cohorts while truncating high-temperature tails.

Scenario 3: Richer Tax Function



$\tau_t = \vartheta_0 + \vartheta_E E_t + \vartheta_\kappa \kappa_t + \vartheta_{TP}(1 - D_{TP})$: the tax responds to carbon intensity and tipping proximity, providing tipping-risk insurance.

The Key Economic Insight

Main result

Pareto-improving policies work through disaster-risk reduction (tail insurance), not maximum mean abatement
(empirical result from Kübler, Scheidegger, Surbek 2025; not proven here).

What the method delivers

- Globally optimal constrained policies
- Transparent, implementable tax rules
- Welfare decomposition by cohort
- Sensitivity to model parameters

What drives the result

- Transfer design resolves intergenerational conflict
- Linear rules capture most of the welfare gain
- Richer rules add tipping-risk insurance
- Simple rules are robust and interpretable

Key Takeaways from Lecture 09

1. **OLG climate models are tractable with DEQNs**: 12 generations + climate + stochastic tipping solved globally.
2. **Pseudo-states enable policy search**: tax parameters + transfer shares as inputs $\rightarrow$ solve once, optimize everywhere.
3. **Pareto-improving policies are computationally feasible**: GP surrogate + constrained optimization in minutes.
4. **Time as state is essential**: nonstationary dynamics (carbon intensity, abatement costs) require t in the neural net input.
5. **Tail insurance, not mean abatement**: the main welfare channel of constrained optimal policy.

Next

Final wrap-up: course synthesis, decision guide, and research frontier.

Notebook Roadmap for Lecture 09

The slides here build on the deterministic CDICE-DEQN solver developed in Lecture 08; the natural notebook to revisit and extend is:

- `lectures/day8/code/03_Stochastic_DICE_DEQN.ipynb` – adds an AR(1) productivity shock to the Lecture-08 deterministic CDICE-DEQN, evaluates the Euler expectation with Gauss–Hermite quadrature, and produces the SCC fan charts that motivate today's UQ discussion (parameter, stochastic, and epistemic uncertainty).

The full OLG-IAM with constrained Pareto-improving tax rules (Kübler, Scheidegger, Surbek 2025, arXiv:2507.01704) is implemented in production code that is not part of the classroom repository. The paper is in the readings folder (`JPE_Macro.pdf`); the classroom notebook focuses on the upstream stochastic CDICE-DEQN building block.

References and Reading Guide

Core references for this lecture

- Kübler, Scheidegger, Surbek (2025). *Using Machine Learning to Compute Constrained Optimal Carbon Tax Rules.* arXiv:2507.01704.
- Kotlikoff, Kubler, Polbin, Sachs, Scheidegger (2021): OLG with carbon-risk taxation.
- Dietz (2024), van der Ploeg and Rezai (2026), Hassler, Krusell, Smith (2016): broader climate-economics background.
- Azinovic, Gaegauf, Scheidegger (2022): DEQN methodology.
- Rasmussen and Williams (2006): Gaussian processes.

File in this repository

- `lectures/day8/readings/JPE_Macro.pdf`

Questions before wrap-up?